

Machine Learning for Statistics and Quantitative Finance

Worksheet: Naive Bayes Classification

Postgraduate Level

Learning Objective

Apply the Naive Bayes classification algorithm manually to different categorical datasets. For each problem, calculate:

Prior $\rightarrow$ Likelihoods $\rightarrow$ Class Scores $\rightarrow$ Prediction

Naive Bayes Classification Rule

For an observation

$$X = (X_1, X_2, \dots, X_p),$$

the Naive Bayes class score for class C_k is

$$S_k = P(C_k) \prod_{j=1}^p P(X_j | C_k)$$

The predicted class is

$$\hat{C} = \arg \max_{C_k} S_k$$

Under the Naive Bayes assumption,

$$P(X_1, \dots, X_p | C_k) = \prod_{j=1}^p P(X_j | C_k)$$

Important

For these exercises, compare the class scores directly. There is no need to calculate the common denominator $P(X)$.

Question 1: Loan Approval

A bank wants to classify a loan application as

$$C_1 = \text{Approved} \quad \text{or} \quad C_2 = \text{Rejected}.$$

The following small teaching dataset contains categorical information about previous loan applicants.

Applicant	Education	Credit History	Loan Status
1	Graduate	Good	Approved
2	Graduate	Good	Approved
3	Graduate	Poor	Rejected
4	Undergraduate	Good	Approved
5	Undergraduate	Poor	Rejected
6	Graduate	Good	Approved
7	Undergraduate	Good	Rejected
8	Graduate	Poor	Rejected
9	Undergraduate	Good	Approved
10	Graduate	Good	Approved
11	Undergraduate	Poor	Rejected
12	Graduate	Poor	Rejected

The two features are:

$$X_1 = \text{Education}, \quad X_2 = \text{Credit History}.$$

The target variable is:

$$C_1 = \text{Approved}, \quad C_2 = \text{Rejected}.$$

Prediction Task

Consider a new applicant with

Education = Graduate

and

Credit History = Good

Use Naive Bayes to predict the loan status.

Step 1: Calculate Prior Probabilities

Calculate:

$$P(\text{Approved})$$

and

$$P(\text{Rejected}).$$

Step 2: Calculate Likelihoods

Calculate:

$$P(\text{Graduate} \mid \text{Approved})$$

$$P(\text{Good} \mid \text{Approved})$$

$$P(\text{Graduate} \mid \text{Rejected})$$

$$P(\text{Good} \mid \text{Rejected}).$$

Step 3: Calculate Class Scores

Calculate:

$$S_{\text{Approved}} = P(\text{Approved})P(\text{Graduate} \mid \text{Approved})P(\text{Good} \mid \text{Approved})$$

and

$$S_{\text{Rejected}} = P(\text{Rejected})P(\text{Graduate} \mid \text{Rejected})P(\text{Good} \mid \text{Rejected}).$$

Step 4: Make the Prediction

Compare the two class scores and report:

$\hat{C} =$ _____

Question 2: Email Spam Classification

An email service wants to classify an email as

$$C_1 = \text{Spam} \quad \text{or} \quad C_2 = \text{Not Spam}.$$

Consider the following small categorical dataset.

Email	Contains “Offer”	Known Sender	Class
1	Yes	No	Spam
2	Yes	No	Spam
3	Yes	Yes	Spam
4	No	No	Spam
5	Yes	No	Spam
6	No	Yes	Not Spam
7	No	Yes	Not Spam
8	Yes	Yes	Not Spam
9	No	Yes	Not Spam
10	No	No	Not Spam
11	Yes	No	Spam
12	No	Yes	Not Spam

The two features are:

$$X_1 = \text{Contains “Offer”}, \quad X_2 = \text{Known Sender}.$$

The target variable is:

$$C_1 = \text{Spam}, \quad C_2 = \text{Not Spam}.$$

Prediction Task

A new email has the following characteristics:

Contains “Offer” = Yes

and

Known Sender = No

Use Naive Bayes to classify this email.

Step 1: Calculate Prior Probabilities

Calculate:

$$P(\text{Spam})$$

and

$$P(\text{Not Spam}).$$

Step 2: Calculate Likelihoods

Calculate:

$$P(\text{Offer} = \text{Yes} \mid \text{Spam})$$

$$P(\text{Known Sender} = \text{No} \mid \text{Spam})$$

$$P(\text{Offer} = \text{Yes} \mid \text{Not Spam})$$

$$P(\text{Known Sender} = \text{No} \mid \text{Not Spam}).$$

Step 3: Calculate Class Scores

Calculate:

$$S_{\text{Spam}} = P(\text{Spam})P(\text{Offer} = \text{Yes} \mid \text{Spam})P(\text{Known Sender} = \text{No} \mid \text{Spam})$$

and

$$S_{\text{Not Spam}} = P(\text{Not Spam})P(\text{Offer} = \text{Yes} \mid \text{Not Spam})P(\text{Known Sender} = \text{No} \mid \text{Not Spam}).$$

Step 4: Make the Prediction

Compare the two class scores:

S_{Spam}	vs.	$S_{\text{Not Spam}}$
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Therefore,

$\hat{C} =$ _____

Question 3: Mushroom Classification

Mushroom classification is a well-known categorical classification problem. The UCI Mushroom dataset contains categorical physical characteristics and a binary target indicating edible or poisonous mushrooms.

For this worksheet, use the following small teaching dataset.

Mushroom	Odor	Cap Color	Class
1	None	Brown	Edible
2	None	Brown	Edible
3	Almond	White	Edible
4	None	White	Edible
5	Almond	Brown	Edible
6	Foul	White	Poisonous
7	Foul	Brown	Poisonous
8	Foul	White	Poisonous
9	None	Brown	Poisonous
10	Foul	Brown	Poisonous
11	Almond	White	Edible
12	Foul	White	Poisonous

The two features are:

$$X_1 = \text{Odor}, \quad X_2 = \text{Cap Color}.$$

The target variable is:

$$C_1 = \text{Edible}, \quad C_2 = \text{Poisonous}.$$

Prediction Task

A new mushroom has:

Odor = None

and

Cap Color = Brown

Use Naive Bayes to classify the mushroom.

Step 1: Calculate Prior Probabilities

Calculate:

$$P(\text{Edible})$$

and

$$P(\text{Poisonous}).$$

Step 2: Calculate Likelihoods

Calculate:

$$P(\text{None} \mid \text{Edible})$$

$$P(\text{Brown} \mid \text{Edible})$$

$$P(\text{None} \mid \text{Poisonous})$$

$$P(\text{Brown} \mid \text{Poisonous}).$$

Step 3: Calculate Class Scores

Calculate:

$$S_{\text{Edible}} = P(\text{Edible})P(\text{None} \mid \text{Edible})P(\text{Brown} \mid \text{Edible})$$

and

$$S_{\text{Poisonous}} = P(\text{Poisonous})P(\text{None} \mid \text{Poisonous})P(\text{Brown} \mid \text{Poisonous}).$$

Step 4: Make the Prediction

Compare:

$$S_{\text{Edible}} \quad \text{and} \quad S_{\text{Poisonous}}$$

Therefore,

$$\hat{C} = \underline{\hspace{2cm}}$$

Common Working Template

Use the following structure for all three questions.

Step 1: Prior Probabilities

$$P(C_1) = \frac{\text{Number of observations in } C_1}{\text{Total number of observations}}$$

$$P(C_2) = \frac{\text{Number of observations in } C_2}{\text{Total number of observations}}$$

Step 2: Likelihoods

For every observed feature, calculate

$$P(X_j \mid C_k) = \frac{\text{Count of } X_j \text{ within class } C_k}{\text{Total count of class } C_k}.$$

Step 3: Class Scores

For each class,

$$S_k = P(C_k) \prod_{j=1}^p P(X_j \mid C_k)$$

Step 4: Prediction

Choose the class with the largest score:

$$\hat{C} = \arg \max_{C_k} S_k$$

Conceptual Questions

Answer briefly.

1. In each of the three problems, what represents the **class variable**?
2. In each problem, what are the **features**?
3. Why do we multiply the likelihoods in Naive Bayes?

4. What does the “Naive” assumption mean?
5. Why can we compare class scores without calculating the denominator $P(X)$?

Final Summary

For every dataset, remember:



$$S_k = P(C_k) \prod_{j=1}^p P(X_j | C_k)$$

$$\hat{C} = \arg \max_{C_k} S_k$$

Calculate $\rightarrow$ Compare $\rightarrow$ Classify